```
import matplotlib.pyplot as plt
import numpy as np

specimens = [
    "Specimen 1\n(Gap 3.0 mm)",
    "Specimen 2\n(Gap 3.0 mm)",
    "Specimen 3\n(Gap 2.3 mm)",
    "Specimen 4\n(Gap 2.3 mm)",
]

# Densities in g/cm³ (based on solid PLA = 1.24 g/cm³)
# Theoretical Apparent Density = (1 - Theoretical Porosity) * 1.24 g/cm³
calc_density_gcm3 = [
    (1 - 0.7368) * 1.24,
    (1 - 0.7368) * 1.24,
    (1 - 0.6820) * 1.24,
    (1 - 0.6820) * 1.24,
]

# Printed Apparent Density = (1 - Apparent Porosity) * 1.24 g/cm³
app_density_gcm3 = [
    (1 - 0.7364) * 1.24,
    (1 - 0.7362) * 1.24,
    (1 - 0.6680) * 1.24,
    (1 - 0.6718) * 1.24,
```

```
]

x = np.arange(len(specimens))
bar_width = 0.28

fig, ax = plt.subplots(figsize=(9, 5.5), dpi=300)

rects1 = ax.bar(
    x - bar_width / 2 - 0.02,
    calc_density_gcm3,
    bar_width,
    label="Calculated Theoretical Density",
    color="#2b5c8f",
    edgecolor="black",
    linewidth=0.8,
)

rects2 = ax.bar(
    x + bar_width / 2 + 0.02,
    app_density_gcm3,
    bar_width,
    label="Apparent Printed Density",
    color="#d95f02",
    edgecolor="black",
    linewidth=0.8,
```

```
)



def add_labels(rects):
    for rect in rects:
        height = rect.get_height()
        ax.annotate(
            f"{height:.3f}",
            xy=(rect.get_x() + rect.get_width() / 2, height),
            xytext=(0, 5),
            textcoords="offset points",
            ha="center",
            va="bottom",
            fontsize=9,
            fontweight="bold",
        )



add_labels(rects1)
add_labels(rects2)

ax.set_title(
    "Theoretical vs. Printed Apparent Density",
    fontsize=11,
    fontweight="bold",
```

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    pad=15,
)
ax.set_ylabel(
    "Apparent Density (g/cm³)", fontsize=11, fontweight="bold"
)
ax.set_xticks(x)
ax.set_xticklabels(specimens, fontsize=9.5, fontweight="bold")
ax.set_ylim(0, 0.55)

ax.set_axisbelow(True)
ax.yaxis.grid(True, linestyle="--", alpha=0.5)
ax.legend(
    loc="upper center",
    bbox_to_anchor=(0.5, -0.15),
    ncol=2,
    frameon=True,
    fontsize=9.5,
)

plt.tight_layout()
plt.show()
```